

Demonstrating an Auditory-Cued Archery Social Exertion Game for the Blind and Sighted to Play Together

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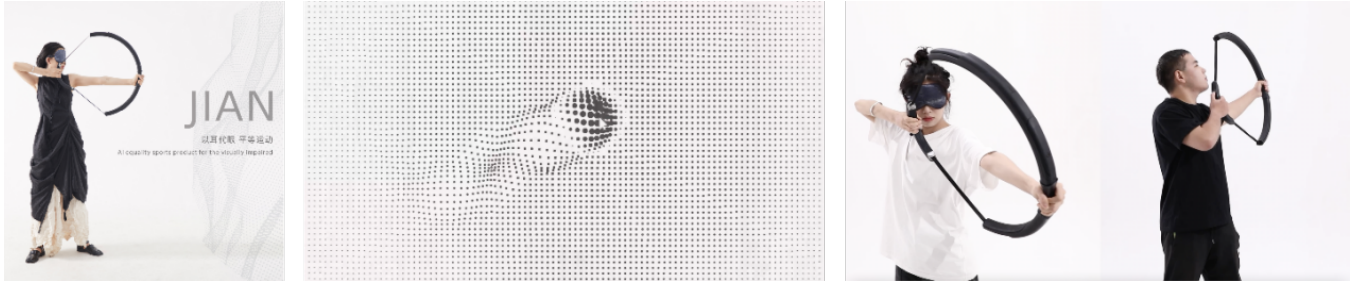


Figure 1: Hearing the Bullseye: an archery social exertion game. Using a bow equipped with an infrared sensor, this game allows both the blind and sighted players to target an unseen bullseye using auditory and tactile feedback, creating a fair environment for the blind.

ABSTRACT

Positive social connection and interaction are crucial for psychological health, especially for people who are blind due to social isolation. Research literature shows that social exertion gameplay can be an effective form of social interaction and can greatly improve individuals' relationships with their friends. But how can the blind play with the sighted? The disparity in visual abilities poses challenges for fairness in social exergames, often lowering the confidence of the blind and causing boredom for the sighted. To overcome these challenges, We demonstrate "Hearing the Bullseye," an archery social exertion game. Using a bow equipped with an infrared sensor, this game allows both the blind and sighted players to target an unseen bullseye using auditory and tactile feedback, creating a fair environment for the blind. Meanwhile, sighted players receive spoiler-free visual feedback to keep them engaged without boredom. The empirical studies show this game strategically creates an inclusive environment and harmonious relationships between the blind and their sighted family and friends through its game design.

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CCS CONCEPTS

• **Human-centered computing** → **Accessibility theory, concepts and paradigms; Interaction design theory, concepts and paradigms; Interaction paradigms.**

KEYWORDS

Accessibility, Exertion Game, Visually Impaired, Auditory Feedback, Archery, Social Play

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1 INTRODUCTION

Providing a positive social connection and interaction for people who are blind and their sighted friends is crucial for their psychological health. Social play [17] serves as an important channel for positive communication between them [8, 20]. Many scholars have emphasized the positive aspects of social activities such as sports [1–3], hobbies [6], family life, playing games [4, 15], and watching shows [11], all of which promote social interaction between the blind and sighted.

Can blind and sighted individuals compete fairly while engaging together? Due to the disparity in visual abilities, fair competition during play is challenging [5, 18, 19], often undermining the confidence of blind participants [21]. Sighted individuals often play an assistant role without fully enjoying the game, leading to disinterest and boredom. Few previous social exertion play studies address the

fairness gap caused by inherent visual ability asymmetry while also preventing sighted individuals from becoming bored. This makes it difficult for both blind and sighted individuals to achieve a positive social experience [5, 14].

We demonstrate "Hearing the Bullseye," an archery social game with auditory cues that enable blind and sighted players to compete fairly. Players use a bow with an infrared sensor to target an unseen bullseye, guided by auditory feedback indicating proximity through changes in sound frequency and tactile feedback via motor vibration. An empirical study with 18 blind and sighted participants showed that both groups thoroughly enjoyed the game. It facilitated fair competition in an inclusive and engaging manner. Therefore, this game strategically creates an inclusive environment and harmonious relationships between the blind and their sighted family and friends through its design.

2 RELATED WORKS

Numerous studies have used assistive technology to enhance social interactions between blind and sighted individuals in game activities. Examples include Sonic-Badminton for physical and social engagement [7]. Other games replace visual feedback with auditory and tactile feedback, such as VI Bowling and VI Tennis [9, 10]. Leisure activities include eyes-free yoga developed with Kinect [13] and a 2D action game for both sighted and blind users [8]. Social experiences encompass enjoying live football matches together [1, 11] and fireworks [16]. Studies also focus on family interaction for blind individuals [12] and emphasize sports and sound interfaces accessible to blind users [14].

3 DESIGN

We designed an archery social exertion game aimed at creating a fair and engaging experience for both sighted and blind players. The game includes an archery bow that provides real-time audio and tactile feedback. As players get closer to aim the target, the audio frequency increases, helping them precisely locate the target center. As the tension of the bowstring increases, the vibrations in the bow handle intensify, offering both auditory and tactile feedback. When the arrow is released at the highest audio frequency, players receive visual feedback for the sighted and auditory feedback for the both indicating success or failure.

The game features both single-player and two-player modes to enhance fairness and engagement. In single-player mode, players aim for the target center based on audio frequency, completing the interaction at the highest frequency. In two-player mode, the game emits two different audio cues, requiring players to identify their own within the game environment. Victory goes to the player who hits their target most frequently. The game aims to let sighted players experience the challenges blind individuals face in distinguishing sounds, while fostering inclusivity and communication between blind and sighted players.

4 USER STUDY

Our user study was conducted during the Alibaba UCAN Design Week, involving over 18 blind and sighted players.

An indoor space measuring 8x8 meters was set up, with the walls and floor used as projection surfaces for the game. The area



Figure 2: Hearing the Bullseye level and two-player mode working example: (a) First level; (b) The second pass; (c) The third pass; (d) Two-player mode.



Figure 3: Photos from the "Hearing the Bullseye" user experiments

included designated spaces for spectators and was outfitted with immersive projectors and audio equipment. Multiple cameras were used to capture player experiences and audience reactions.

First, we explained the game rules. Both sighted and blind players learned the game controls through live demonstrations, verbal instructions, and physical actions. Sighted players received a 5-minute tutorial, while blind players were instructed using tactile methods and sound cues.

Next, we started game testing. The testing involved both single and dual-player modes. Participants completed three levels of increasing difficulty, with shooting accuracy data recorded. In two-player matches, blind and sighted players were randomly paired until one hit the target.

Finally, post-game feedback was collected through semi-structured interviews, lasting between 20 to 30 minutes, focusing on players' overall experiences, perceptions of fairness, and reactions to the game's effects.

5 FINDINGS

Fairness for Engagement. Both blind and sighted participants reported high levels of engagement and enjoyment. The fair competitive elements kept players interested and motivated throughout the sessions. Blind players found the audio cues intuitive and crucial for gameplay, while sighted players valued the visual feedback. Both blind and sighted participants reported high levels of engagement and enjoyment. The fair competitive elements kept players interested and motivated throughout the sessions, with blind players often excelling due to their reliance on sound.

Spoiler-free visual feedback for Sighted. After hitting or missing the target, we will provide strong, spoiler-free visual feedback. Studies show that this strategy effectively keeps the audience engaged.

The cheers from sighted spectators Successfully hitting the target in the game earned blind players enthusiastic cheers from the sighted audience surrounding the arena, validating their efforts. Feedback from the blind players shows that confirmation from spectators is stronger than audio feedback for confirming hits. Both players and spectators are actively engaged in the experience.

6 CONCLUSION

Our archery social exertion game created a more harmonious and friendly play experience for both blind and sighted participants. It allowed blind individuals to engage and compete fairly with friends. The game boosted blind participants' self-confidence and deepened sighted individuals' understanding of the blind, fostering an inclusive environment and promoting harmonious relationships between blind individuals and their sighted friends.

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